

REACTION WHEEL ATTITUDE CONTROL SYSTEM FOR NANO SATELLITES

Personal Engineering Project Report

Design, Modelling and Structural-Dynamic Validation of a CubeSat-Class Reaction Wheel Attitude
Control System

Tools used: Creo Parametric 11.0 · MSC ADAMS · ANSYS Mechanical

Domain: Aerospace / Mechanical Engineering — Attitude Determination & Control Systems (ADCS)

JAN'26 – MAR'2026

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Abstract

Nano satellites, particularly CubeSats, have gained significant importance in modern space missions due to their compact size, low cost, and rapid development capabilities. However, one of the major challenges associated with nano satellites is achieving precise attitude control under strict constraints of size, mass, and limited power availability. Efficient orientation control is essential for mission-critical operations such as communication, Earth observation, and power generation.

To address this challenge, this project proposes a Reaction Wheel-based Attitude Control System (ACS), which provides a compact and efficient solution for controlling satellite orientation. The system operates on the principle of conservation of angular momentum, where the rotation of a flywheel generates an equal and opposite reaction torque on the satellite, enabling precise attitude adjustments.

In this project, a reaction wheel system consisting of a flywheel, electric motor, and supporting structure has been designed. The flywheel is dimensioned to achieve sufficient moment of inertia while maintaining low mass and compact size. A 3D CAD model of the complete assembly has been developed using Creo Parametric 11.0 to ensure proper alignment, balance, and structural feasibility. Additionally, the system design considers integration with control components such as microcontrollers, sensors, and motor drivers for future implementation.

The proposed system aims to provide smooth, continuous, and accurate attitude control while satisfying the constraints of nano satellites. The design demonstrates feasibility and scalability, making it suitable for further development into a functional prototype or simulation-based validation.

1. Introduction

Nano satellites are a class of small satellites with a mass typically ranging from 1 kg to 10 kg. Among them, CubeSats have emerged as one of the most widely adopted platforms due to their standardized structure (typically $10 \times 10 \times 10$ cm units), low development cost, and shorter development cycles. These satellites are extensively used in applications such as Earth observation, communication systems, scientific experiments, and technology demonstration missions.

The rapid growth of nano satellite missions is driven by advancements in miniaturized electronics and the availability of commercial off-the-shelf (COTS) components. However, despite their advantages, nano satellites operate under strict constraints in terms of size, mass, and power. Typically, a CubeSat generates less than 3 W of power, which significantly limits the design and implementation of onboard subsystems.

One of the most critical subsystems of any satellite is the Attitude Determination and Control System (ADCS). The primary function of ADCS is to control and maintain the orientation (attitude) of the satellite in space. Proper orientation is essential to ensure that antennas are directed toward ground stations, sensors are accurately aligned with targets, and solar panels receive maximum sunlight for power generation.

In the space environment, satellites are subjected to various external disturbances such as gravity gradient forces, atmospheric drag (in low Earth orbit), and solar radiation pressure. These disturbances can cause unwanted rotation and instability. Without an effective attitude control system, the satellite may lose its orientation, leading to mission failure.

Traditionally, large satellites use complex and expensive attitude control systems. However, such systems are not suitable for nano satellites due to their size, cost, and power limitations. Therefore, compact and efficient control mechanisms are required.

Among the various attitude control methods, reaction wheels are widely used in nano satellites due to their ability to provide precise, smooth, and continuous control. A reaction wheel is a rotating mass driven by an electric motor. By accelerating or decelerating the wheel, torque is generated, which changes the orientation of the satellite based on the principle of conservation of angular momentum.

Conventional attitude control systems used in larger satellites are often bulky, expensive, and consume significant power. These characteristics make them unsuitable for nano satellite applications, where strict constraints on size, mass, and power must be satisfied. Additionally, earlier nano satellites relied on passive control methods, which offer limited accuracy and cannot meet the requirements of advanced missions such as high-resolution imaging and precise communication.

Therefore, there is a need to develop a compact, lightweight, and energy-efficient attitude control system that can provide accurate and reliable orientation control within the constraints of nano satellites. The primary objective of this project is to design and analyze a reaction wheel-based attitude control system suitable for nano satellite applications.

1.1 Objectives

- To study the fundamentals of attitude control systems used in nano satellites
- To understand the working principle of reaction wheels based on conservation of angular momentum

- To design a compact reaction wheel mechanism (flywheel) capable of generating sufficient control torque
- To select appropriate motor and structural components for efficient operation
- To develop a CAD model of the reaction wheel assembly using suitable software tools
- To analyze design parameters such as size, weight, and feasibility within nano satellite constraints

1.2 Basics of Attitude Control

Attitude Determination and Control Systems (ADCS) for nanosatellites frequently utilize reaction wheels as primary actuators to achieve high-precision pointing and stabilization. These devices function as momentum exchange actuators, leveraging the principle of conservation of angular momentum: by accelerating an internal motorized flywheel, an equal and opposite torque is exerted on the satellite body, thereby altering its orientation. For comprehensive 3-axis control, reaction wheels are typically arranged in an orthogonal or tetrahedral configuration to manage the satellite's primary degrees of freedom: roll, pitch, and yaw.

In a standard 3-axis stabilized mission, the reaction wheels are aligned with the satellite's principal axes of inertia. Adjustments to the flywheel's angular velocity generate the necessary control torques to correct errors in roll (rotation about the longitudinal axis), pitch (rotation about the lateral axis), and yaw (rotation about the vertical axis). Because external environmental disturbances, such as atmospheric drag and solar radiation pressure, can cause a persistent build-up of angular momentum, reaction wheels may eventually reach their maximum rotational speed, a state known as saturation. To recover control capability, nanosatellites often employ secondary actuators, such as magnetorquers, to perform momentum dumping — desaturating the wheels by transferring excess momentum to the Earth's magnetic field.

1.3 Reaction Wheel Concept

The operational foundation of a reaction wheel is based on the Law of Conservation of Angular Momentum. In the isolated environment of space, the total angular momentum of a nanosatellite remains constant unless acted upon by an external torque. By integrating a motorized flywheel — the reaction wheel — into the satellite's internal structure, the system can manipulate its orientation by redistributing momentum between the wheel and the satellite body. When the internal motor applies a torque to increase the flywheel's spin velocity, the satellite body experiences an equal and opposite reactive torque. This fundamental interaction allows for precise control over the spacecraft's attitude without the need for expendable propellants.

Basically, total angular momentum of the whole satellite + reaction wheels system is conserved.

$$I_s \times \omega_s + I_{rw} \times \omega_{rw} = 0$$

$$I_{rw} = - (I_s \times \omega_s) / \omega_{rw}$$

This negative sign means that the reaction wheels are to be rotated in the opposite sense to that of the satellite. To maintain stability across all three spatial dimensions, a nanosatellite typically employs a minimum of three wheels aligned with the roll, pitch, and yaw axes. By precisely modulating the speed of each wheel, the control system can counteract internal vibrations or rotate the vehicle to a specific target orientation.

2. Literature Review

The growing proliferation of nano-satellites — small spacecraft typically weighing between 1 kg and 10 kg — has transformed the landscape of modern space exploration and Earth observation. Despite their compact form, these platforms must perform highly precise orientation tasks such as Earth imaging, inter-satellite communication, and scientific data acquisition — all of which depend critically on accurate Attitude Determination and Control Systems (ADCS). Reaction wheels are the most widely employed actuators for three-axis attitude control in nano-satellites.

The theoretical backbone of any reaction wheel system is the conservation of angular momentum. Markley and Crassidis (2014), in *Fundamentals of Spacecraft Attitude Determination and Control*, provide a comprehensive mathematical framework describing how reaction wheels exchange angular momentum with a spacecraft body. As the wheel accelerates or decelerates, Newton's Third Law dictates a counter-rotation of the satellite proportional to the inertia ratio between the wheel and the spacecraft.

Wertz (1978), in *Spacecraft Attitude Determination and Control*, expanded this foundation to multi-axis systems, demonstrating that a minimum of three reaction wheels — mounted along orthogonal axes — are required for full three-axis control. This work formalised how external torques alter the total system angular momentum while internal torques (such as wheel spin-up) redistribute momentum between the wheel and spacecraft without net external effect.

The mechanical design of the flywheel is the most direct determinant of a reaction wheel's angular momentum storage capability and torque output. The primary design problem is the optimisation of flywheel geometry to maximise the inertia-to-mass ratio within the geometric envelope imposed by the satellite form factor. Oland and Schlanbusch (2009), in *Reaction Wheel Design for CubeSats*, present a systematic sizing methodology applicable to CubeSat missions, deriving design equations that relate flywheel dimensions to achievable angular momentum, and identifying operational altitudes at which magnetic torquers are suitable for momentum dumping — a consideration directly linked to flywheel saturation design.

From a commercial design perspective, research on a miniaturised flight-model reaction wheel only 33 mm in diameter and weighing under 38 g demonstrates that COTS brushless DC motors paired with appropriately sized flywheels can meet CubeSat ACS requirements. Torque-versus-diameter studies mapping state-of-the-art reaction wheel models indicate that wheels below 6 cm diameter are primarily suited to CubeSat applications, while larger satellites require larger flywheels and more powerful motors.

2.1 PID-Based Control as the Standard Approach

Most nanosatellite RWACS implementations use PID controllers due to their simplicity and reliability. Studies such as Narkiewicz et al. (2020) and Long (2014) demonstrate successful implementation across detumbling, nominal control, and momentum unloading, including real hardware validation with IMU and sensor fusion.

2.2 Multi-Loop and Structured PID Architectures

Advanced PID configurations such as dual-loop control (an inner loop for wheel speed and an outer loop for attitude) improve system stability. This has been validated using transfer function modelling, Bode plots, and step response analysis for all three axes.

2.3 Sensor-Based Feedback and Actuation Loop

Gyroscopic sensors (IMUs) are used to measure angular velocity and detect disturbance-induced attitude errors. This data is processed by an onboard microcontroller, which computes the control action (e.g. via PID) and sends torque commands to the reaction wheel motors to generate stabilizing motion. Standard ADCS design frameworks describe this sensor–controller–actuator loop as the core of satellite attitude stabilization.

2.4 Optimized and Intelligent Control Techniques

Researchers have enhanced PID performance using optimization methods such as Genetic Algorithms, achieving better rise time and reduced steady-state error. Additionally, fuzzy logic controllers are used for robustness against uncertainties and efficient momentum management.

3. Problem Formulation

3.1 Context: The Role of Precision Attitude Control

Reaction wheels (RW) serve as the primary actuators for high-precision, propellant-free attitude control in modern satellite missions. By modulating the angular momentum of an internal flywheel, these systems generate predictable reaction torques that enable three-axis stabilization. This capability is critical for missions requiring high pointing stability, such as Earth observation, deep-space telemetry, and orbital telescope positioning.

3.2 The Identified Gap: Theoretical vs. Dynamic Disparity

A review of existing literature reveals significant reliance on idealized mathematical models to predict maneuver performance. Most references prioritize the conservation-of-angular-momentum equations but fail to account for the non-linearities and parasitic effects inherent in a physical mechanism. Specifically, there is a notable absence of integrated multi-body dynamics (MBD) and structural analysis to validate how mechanical imbalances, bearing friction, and structural deformations affect actual slew-maneuver accuracy.

3.3 Problem Statement

There is a need to develop a comprehensive design-to-validation framework for a compact reaction wheel system that goes beyond purely theoretical estimates. While the basic physics of a 30° maneuver can be calculated manually, actual system performance is constrained by:

- Structural Integrity — how the flywheel responds to high centrifugal loads (ANSYS).
- Dynamic Fidelity — how motor torque ripples and internal friction impact settling time and jitter (ADAMS).
- Geometric Optimization — how to achieve maximum inertia within the strict mass and volume envelope of a nano-satellite.

3.4 Technical Objectives

The objective of this project is to synthesize a robust design-and-analysis pipeline that bridges the gap between theory and hardware:

- Synthesis — perform high-fidelity theoretical calculations to size the flywheel (radius, mass, thickness) based on target slew rates and motor RPM limits.
- Virtual Prototyping — construct a precise CAD model in Creo Parametric to define the system's mass properties and centre of gravity.
- Structural Validation — conduct Finite Element Analysis (FEA) in ANSYS to ensure the mechanism can withstand rotational stresses without deformation.
- MBD Verification — simulate the complete maneuver in MSC ADAMS to quantify the difference between the idealized maneuver time and the physical settling time, validating the control logic under realistic dynamic conditions.

4. Theory & Working Principle

4.1 Control Strategy

The control system of a Reaction Wheel Attitude Control System (RWACS) is a closed-loop feedback system designed to regulate and maintain the orientation (attitude) of a satellite in space. The system operates on the principle of conservation of angular momentum, where a change in the angular velocity of the reaction wheel produces an equal and opposite reaction torque on the satellite body.

The control process begins with attitude determination, where onboard sensors such as gyroscopes, inertial measurement units (IMUs), and magnetometers measure the current angular velocity and orientation of the satellite. These measurements are compared with the desired reference attitude to generate an attitude error signal.

This error signal is processed by a control algorithm, most commonly a Proportional–Integral–Derivative (PID) controller, which computes the required control torque. The PID controller combines proportional, integral, and derivative actions to ensure fast response, reduced steady-state error, and system stability. The control output is then converted into motor commands that regulate the angular velocity of the reaction wheels.

The generated torque is related to the reaction wheel dynamics through:

$$T = I_w \cdot \alpha_w$$

where T is the torque, I_w is the moment of inertia of the reaction wheel, and α_w is its angular acceleration. Thus, the control system effectively converts attitude error → control torque → wheel speed variation, ensuring precise attitude stabilization.

5. System Design

5.1 Overall System Architecture

5.1.1 Outer Body Structure (Satellite Frame)

The outer structure of the system is designed as a rigid cubic frame representing the satellite body. It is fabricated using six square plates of mild steel (MS), each of thickness 3 mm.

Key features:

- Material: Mild Steel (MS)
- Thickness: 3 mm
- Configuration: Closed cube (6 faces)
- Provides structural rigidity
- Houses all internal components (reaction wheels, electronics, motor mounts)
- Acts as the main reference body for attitude motion

Reason for selection — mild steel offers:

- Good strength and stiffness
- Ease of machining and welding
- Cost-effectiveness for prototyping

5.1.2 Reaction Wheel Configuration

The system uses three reaction wheels, each aligned along mutually perpendicular axes (X, Y, Z directions) to achieve full 3-axis attitude control.

Wheel specifications:

- Material: Mild Steel
- Thickness: 2 mm
- Diameter: 52 mm
- Number of wheels: 3
- Orientation: Orthogonal (X, Y, Z axes)

Function:

- Each wheel controls rotation about one axis
- By accelerating/decelerating the wheel, torque is generated on the satellite body

5.1.3 Reaction Wheel Assembly

Each reaction wheel is mounted on a BLDC motor shaft and forms a compact actuator unit. The assembly is designed for proper alignment, balance, and secure mounting. Components of the assembly:

- Reaction Wheel (Flywheel) — mounted directly on the motor shaft; provides the required inertia for torque generation

- Threaded Shaft — transfers torque from motor to wheel; ensures a rigid connection
- Collar — maintains axial positioning of the wheel; prevents unwanted axial movement
- Nut (top locking element) — secures the wheel firmly to the shaft; prevents loosening during high-speed rotation
- BLDC Motor — drives the reaction wheel; provides controlled angular velocity
- Mount Plate — fixes the motor to the satellite structure; ensures alignment and stability

5.1.4 BLDC Motor Specifications

The system uses an A2212 1000KV BLDC motor for each reaction wheel.

Electrical specifications:

- KV rating: 1000 RPM/V
- Operating voltage: 7.4 V – 11.1 V (2S–3S LiPo)
- Max current: ~12 A
- Power output: ~150 W

Mechanical specifications:

- Weight: ~55–60 g
- Shaft diameter: 3.17 mm
- Motor diameter: ~27–28 mm
- Motor height: ~30 mm

Reason for selection:

- High speed capability
- Lightweight and compact
- Easily available and cost-effective
- Suitable for rapid prototyping

5.1.5 Electronics and Control Components

The system includes several electronic components for sensing, control, and actuation:

- Microcontroller (e.g. Arduino / STM32) — acts as the main controller; executes control algorithms (PID); processes sensor data and generates control signals
- Motor Driver (ESC – Electronic Speed Controller) — controls speed of the BLDC motor; converts PWM signal into motor rotation; enables precise speed control
- IMU Sensor (gyroscope + accelerometer) — measures angular velocity and orientation; provides real-time feedback for the control system
- Power Supply System — typically a Li-Po battery; supplies power to motors, controller, and sensors
- Voltage Regulator — steps down battery voltage; provides stable voltage to electronics
- Connecting Wires and PCB — ensures electrical connections; organized layout of the circuit

- Mounting Hardware — screws, bolts, spacers used for fixing components inside the structure

5.1.6 Working Integration of Components

- The IMU sensor detects the current orientation
- Data is sent to the microcontroller
- The controller computes the required correction torque
- A signal is sent to the ESC
- The ESC drives the BLDC motor
- The motor rotates the reaction wheel
- Due to conservation of angular momentum, the satellite body rotates in the opposite direction

6. Reaction Wheel Sizing — Design Calculations

The following steps were used to size the reaction wheel parameters. The complete vector equation of rotational motion for the satellite system — the angular momentum conservation equation expanded — is:

$$T_{\text{ext}} = I_s \times \dot{\omega}_s + \omega_s \times I_s \times \omega_s + I_{RW} \times \dot{\omega}_{RW} + \omega_s \times I_{RW} \times \omega_{RW}$$

To simplify the analysis, it is assumed that only one wheel rotates at a time, making the gyroscopic cross-coupling terms zero:

$$T_{\text{ext}} = I_s \times \dot{\omega}_s + I_{RW} \times \dot{\omega}_{RW}$$

In the absence of external torque:

$$I_s \times \omega_s = I_{RW} \times \omega_{RW}$$

The moment of inertia of the satellite is taken from the CAD model using iterative methods. I_s is taken to be the maximum of (I_{xx} , I_{yy} , I_{zz}), which is $I_{yy} = 0.1478619 \text{ kg}\cdot\text{m}^2$. The slew rate (cruising speed, ω_s) can be chosen between $2^\circ/\text{s}$ and $5^\circ/\text{s}$; $\omega_s = 5^\circ/\text{s}$ was selected. The operational speed of the motor is between 8000–12000 RPM; $\omega_{RW} = \omega_{\text{max}}$ at 9000 RPM was taken.

$$0.1478619 \times (5 \times \pi/180) = I_{RW} \times (2\pi \times 9000/60)$$

$$I_{RW} = 1.36909 \times 10^{-5} \text{ kg}\cdot\text{m}^2$$

Fixing the thickness of the reaction wheel at 2 mm and the density of Alloy Steel at $7850 \text{ kg}/\text{m}^3$, the required radius of the reaction wheel is found:

$$I_{RW} = \frac{1}{2} \times M \times R^2 = \frac{1}{2} \times (\rho \times \pi R^2 \times t) \times R^2$$

$$1.36909 \times 10^{-5} = \frac{1}{2} \times 7850 \times \pi \times R^4 \times 0.002$$

$$R = 0.027296 \text{ m} = 27.296 \text{ mm} \quad M = 0.036750 \text{ kg} = 36.750 \text{ g}$$

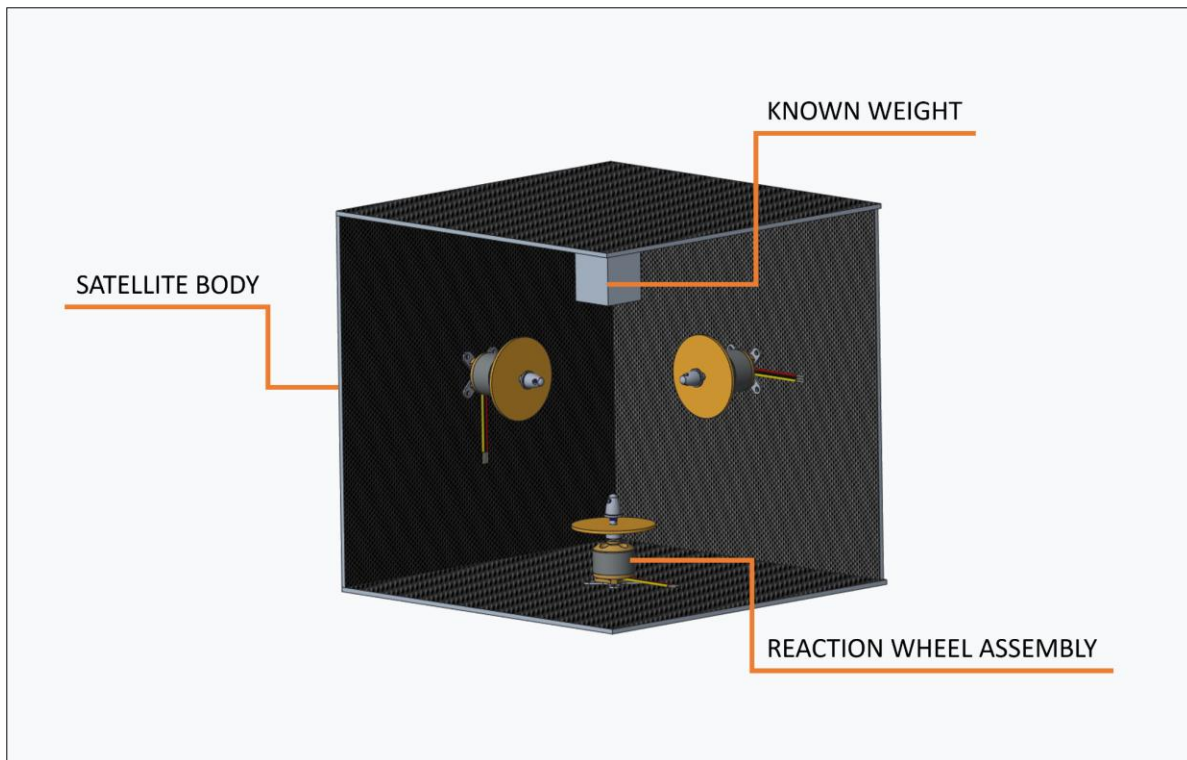
The calculations were repeated in a spreadsheet for several iterations of the input parameters:

I_s (kg·m ²)	ω_s (°/s)	ω_s (rad/s)	RPM	ω_{RW} (rad/s)	I_{RW} (kg·m ²)	t_{RW} (mm)	RRW (mm)	MRW (g)
0.14786	2	0.03490	9000	942.480	5.476×10^{-6}	2	21.708	23.243
0.14786	2	0.03490	5000	523.600	9.857×10^{-6}	2	25.144	31.183
0.14786	2	0.03490	9000	942.480	5.476×10^{-6}	2	21.708	23.243
0.14786	5	0.08726	9000	942.480	1.369×10^{-5}	2	27.296	36.750
0.14786	5	0.08726	5000	523.600	2.464×10^{-5}	2	31.617	49.305
0.14786	5	0.08726	9000	942.480	1.369×10^{-5}	2	27.296	36.750

The final design (4th row above) was carried forward into the CAD model and subsequent analyses.

7. CAD Model

The complete reaction wheel assembly and satellite frame were modelled in Creo Parametric 11.0, based on the dimensions derived from the sizing calculations above.



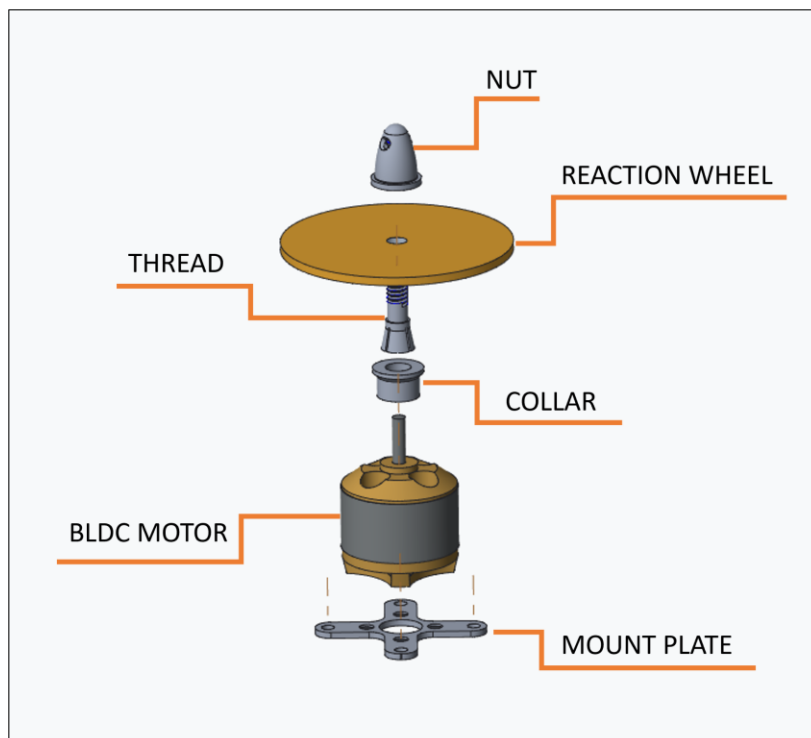
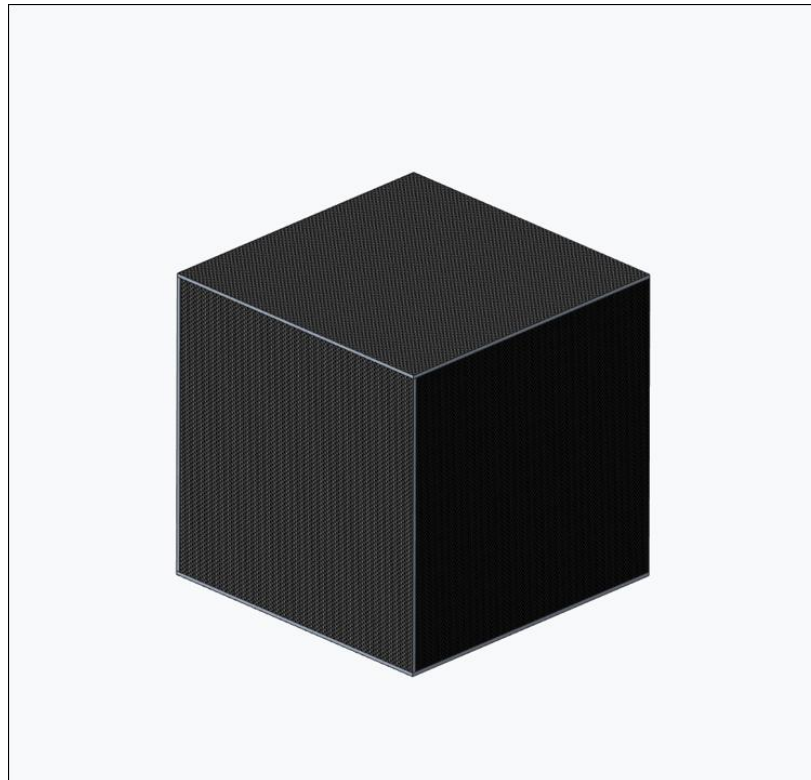


Fig. 7.1 — CAD assembly of the reaction wheel attitude control system

8. Multi-Body Dynamics (MBD) Analysis — MSC ADAMS

8.1 Introduction

The objective of this study is to perform a kinematic analysis of the reaction wheel system using MSC ADAMS, focusing on the relationship between motor input and satellite angular response. The simulation helps in understanding system dynamics such as angular displacement, velocity, acceleration, torque generation, and electrical characteristics like current consumption.

8.2 Geometry and Model Description

The model consists of the following main components:

- Satellite Body (Main Frame) — represented as a rigid body with defined mass and moment of inertia; free to rotate about a specified axis
- Reaction Wheel (Flywheel) — mounted internally on the satellite body; designed as a rotating disc with known mass and radius; high moment of inertia to maximize angular momentum exchange
- Motor Shaft Connection — the reaction wheel is connected to the satellite via a revolute joint allowing rotation about a single axis (typically Z-axis)
- Reference Frame — a global reference frame is defined; satellite motion is measured relative to this frame

Assumptions:

- The system is considered rigid (no deformation)
- External disturbances (gravity gradient, drag, etc.) are neglected
- Only internal torque exchange is considered

8.3 Boundary Conditions and Loading

8.3.1 Boundary Conditions

- The satellite body is either free in space (floating condition) or constrained with a single rotational DOF, depending on the setup
- The reaction wheel is connected via a revolute joint, allowing rotation about one axis only

8.3.2 Initial Conditions

- Initial angular velocity of both satellite and wheel = 0 rad/s
- Initial angular displacement = 0°

8.3.3 Loading Conditions

- No external forces or torques are applied
- Only the internal torque generated by the motor drives the system

8.4 Motor Creation and Settings in MSC ADAMS

The motor is the key driver of the system. It is applied at the revolute joint connecting the reaction wheel and the satellite.

Steps to create the motor:

- Select the revolute joint between the reaction wheel and satellite
- Insert Motion → Rotational Motion
- Define motion type: select Angular Velocity (RPM or rad/s) as input
- Input function (see profile below)

Define a runtime function

```
STEP(time,0,0,2.5,942)+STEP(time,3.5,0,6,-942)
```

Modify Motor

Step 5 of 6

Motor Geometry • **Inputs** • Motor Output

Distribution Factor	0.96	Strength of PM Field (T)	1.2
Coil Pitch Factor	0.87	Length Of the Rotor Windings	30.0
Slot Skew Factor	1.0	Radius Of the Armature	14.0
Teeth per Phase Count	4	Flux Per Pole (Wb)	3.0E-03
No. of Turns per Phase	120	No. of Conductors	300
Armature Resistance (Ohms)	0.15	No. of Paths	2
No. of Poles	14		
Maximum Torque	500.0	Max. Ang. Velocity	1256.0
Control Method	Speed Control		
Enter Target Value	Expression	STEP(time,0,0,2.5,942)+STEP(time,3.5,0,6,-942)	
Friction Torque	0.1	Gain	1.0
Damping Coefficient	1.0E-03	I Gain	1.0
		P Gain	0.1
		D Gain	1.0E-03

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Fig. 8.1 — Motor input function and profile in MSC ADAMS

- Ensure unit consistency (rad/s preferred for analysis)

Motor parameters considered:

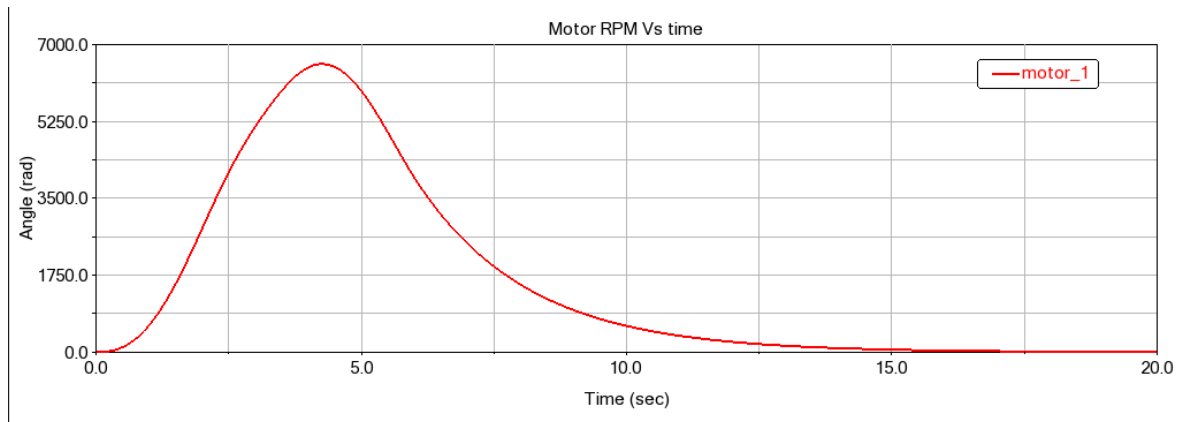
- Maximum angular velocity (e.g. up to 9000–12000 RPM)

- Torque limits based on motor capability
- Time required to reach desired speed

8.5 Simulation Outputs and Results

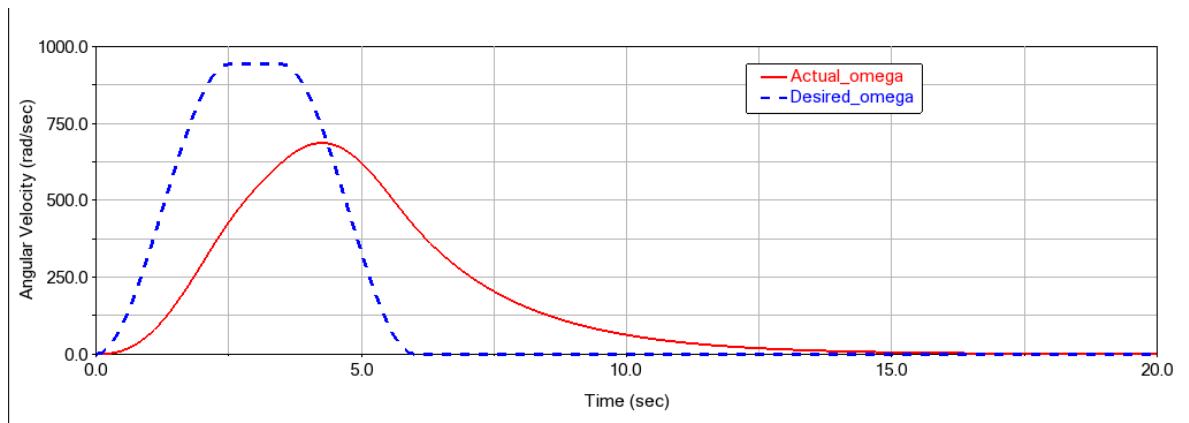
8.5.1 Motor RPM vs Time

Shows how quickly the reaction wheel accelerates and helps verify motor capability against the required speed.



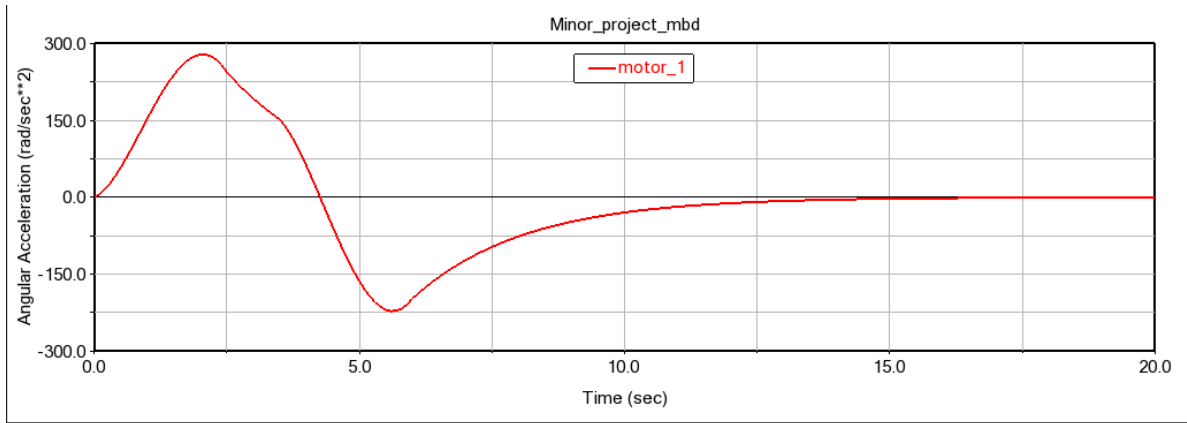
8.5.2 Angular Velocity (Wheel & Satellite)

Reaction wheel velocity increases while the satellite rotates in the opposite direction due to conservation of momentum.



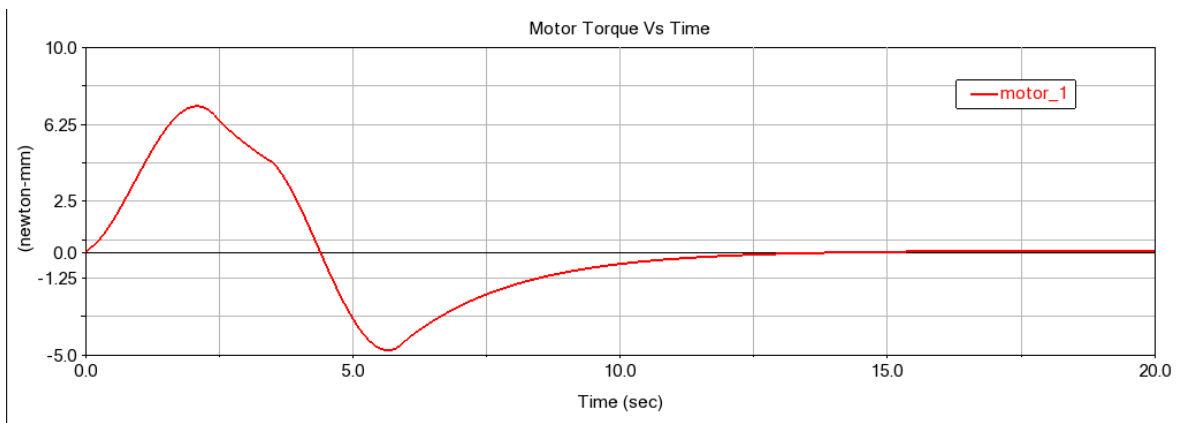
8.5.3 Angular Acceleration

Angular acceleration peaks during motor start-up, indicating torque demand.



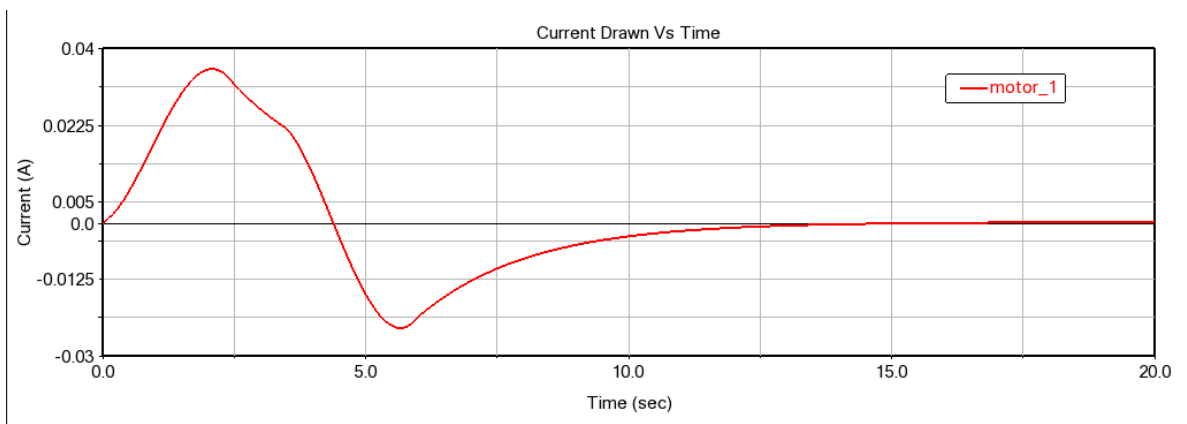
8.5.4 Torque vs Time

Torque is high during the acceleration phase, drops during constant velocity, and reverses during deceleration (if applied).



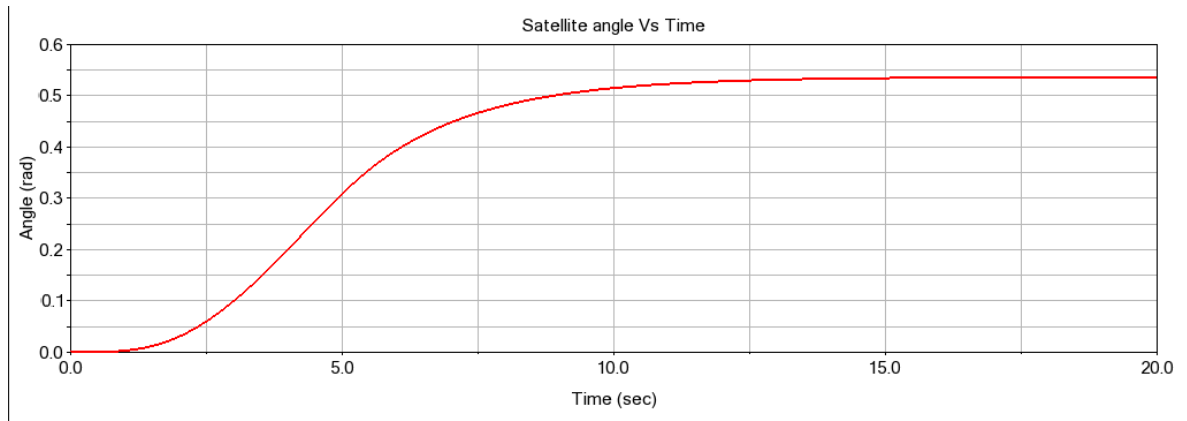
8.5.5 Current Drawn

Current drawn is directly proportional to torque, peaking during the acceleration phase — an important consideration for power system design.



8.5.6 Satellite Angle vs Time

This is the key output for attitude control, confirming whether the desired rotation (e.g. 30° in 11 s) is achieved.



9. Computer-Aided Engineering (CAE) Analysis — ANSYS

9.1 Introduction

CAE analysis was performed to validate the structural integrity and dynamic behaviour of the Reaction Wheel Attitude Control System (RWACS). The primary objective was to ensure that the designed components, particularly the flywheel and housing, can safely withstand operational and launch-induced loads without failure. Special emphasis was given to:

- Avoidance of resonance with the motor excitation frequency
- Structural stability under random vibration (launch conditions)
- Verification of design reliability using standardized aerospace loading conditions

9.2 Geometry and Model Description

The CAD model of the reaction wheel assembly was developed in Creo Parametric and imported into ANSYS Mechanical for analysis. The CAE study includes the flywheel (primary rotating component) and the RWACS housing (supporting structure). The flywheel is designed as a thin circular disc with a central hole, based on the calculated parameters:

- Radius: 27.29 mm
- Thickness: 2 mm
- Mass: 36.75 g

The RWACS housing encloses the motor and flywheel assembly and provides mounting interfaces to the satellite structure.

9.3 Material Properties

The flywheel material is Alloy Steel. The following properties were used in the simulation (assumed linear elastic and isotropic):

Property	Value
Density	7850 kg/m ³
Young's Modulus	2×10^5 MPa
Poisson's Ratio	0.3

9.4 Meshing Details

The geometry was discretized using tetrahedral elements in ANSYS Mechanical.

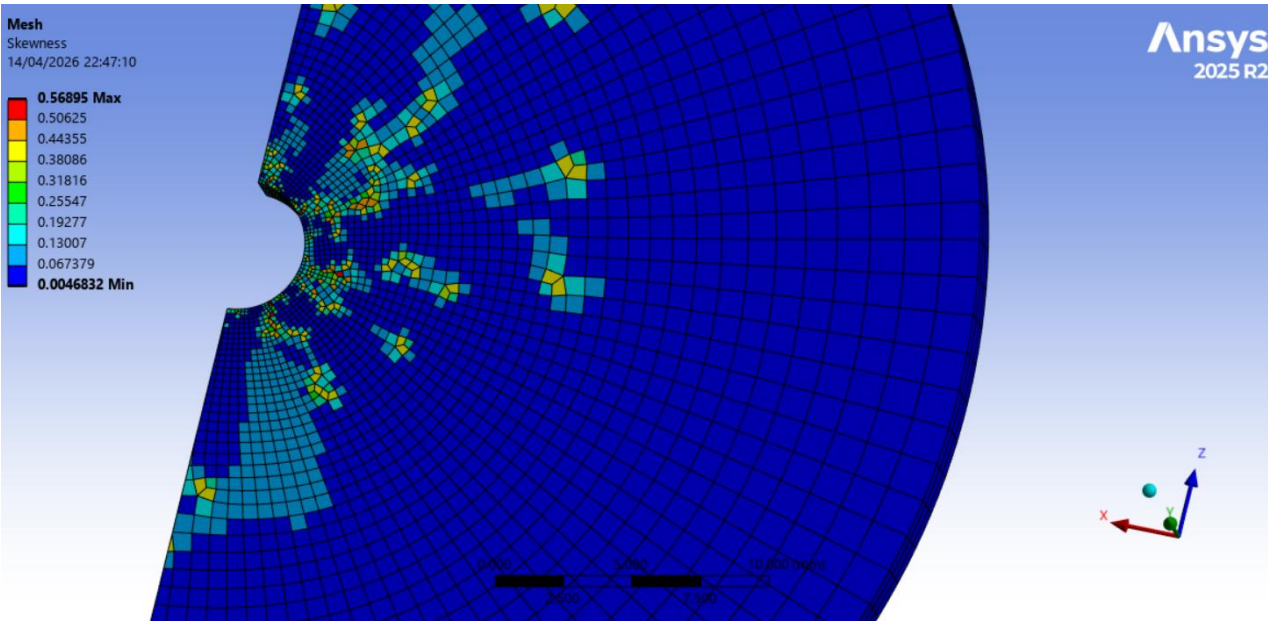
Mesh specifications:

- Element type: Hex & Tetrahedral
- Mesh method: Patch conforming
- Refinement applied at the inner hole region and outer circumference

Mesh quality criteria:

- Skewness < 0.6
- Orthogonality: 0.5 – 0.8
- Tetrahedral collapse: 0.7 – 1

These criteria ensure accurate stress distribution and numerical stability of the solution. A fine mesh was used in critical regions to capture stress concentration and dynamic response accurately.



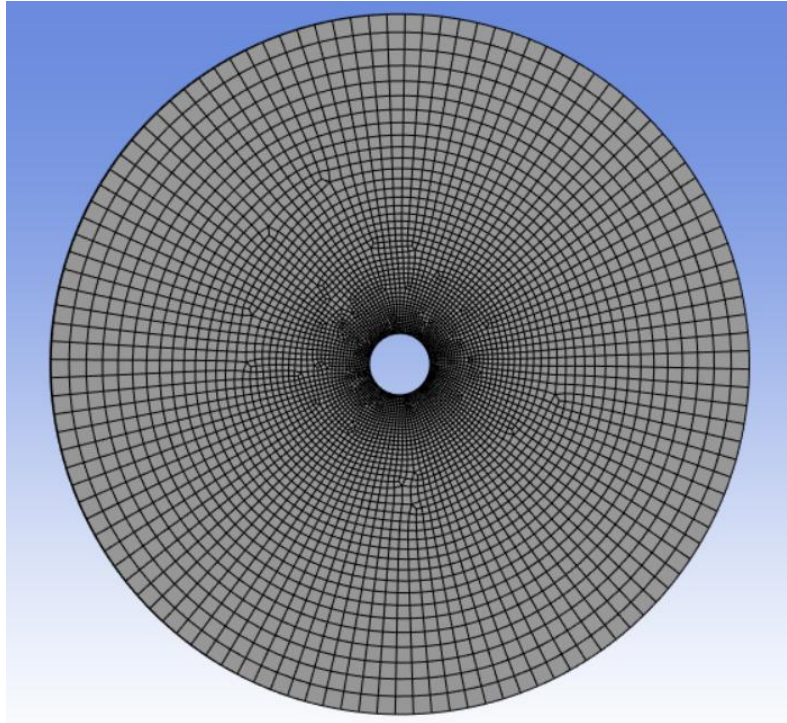


Fig. 9.1 — Flywheel mesh (left) and skewness check (right)

9.5 Boundary Conditions and Loading

9.5.1 Modal Analysis (Pre-Stressed)

To simulate realistic operating conditions, a pre-stressed modal analysis was performed. The mounting holes/regions were assigned fixed support, restricting all degrees of freedom, and a rotational velocity corresponding to the operating speed (9000 RPM) was applied about the axis of rotation. This approach accounts for centrifugal stiffening effects in the rotating flywheel.

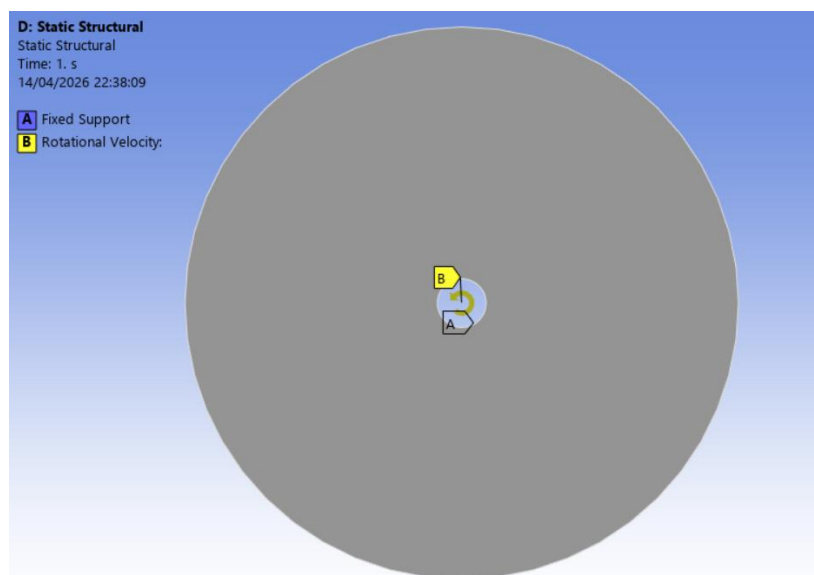
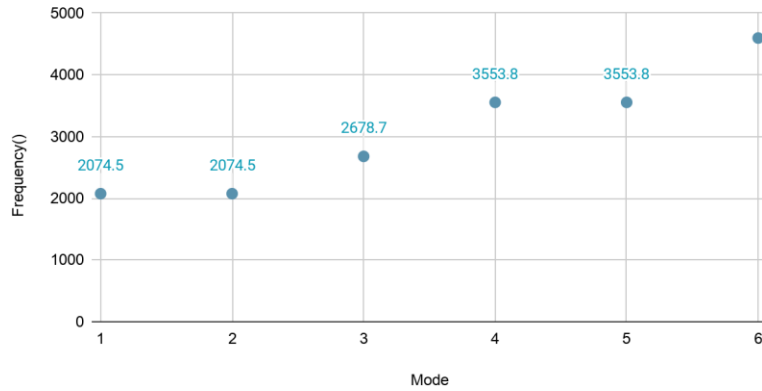


Fig. 9.2 — Boundary condition for pre-stressed modal analysis

Pre Stress Modal Analysis of Flywheel



Mode	Frequency (Hz)
1	2074.5
2	2074.5
3	2678.7
4	3553.8
5	3553.8
6	4595.9

Table 9.1 — First six natural frequencies of the flywheel

9.5.2 Random Vibration Analysis

Random vibration analysis was conducted to simulate launch and environmental conditions.

- Input: Power Spectral Density (PSD) acceleration curve
- Source: General Environmental Verification Specification (GEVS)
- Frequency range: 20 Hz – 2000 Hz

The same PSD curve was consistently used for both the flywheel and the RWACS housing to ensure uniform excitation conditions across all components.

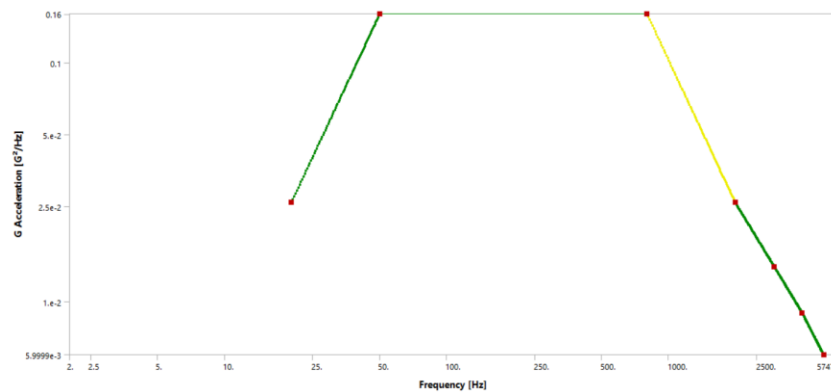


Fig. 9.3 — PSD (G^2 / Hz) acceleration curve used as input

9.6 Solver Settings

- Software: ANSYS Mechanical
- Analysis types: Pre-stressed Modal Analysis; Random Vibration Analysis (PSD)

The modal results were used as input for the random vibration analysis to evaluate system response under dynamic excitation.

10. Implementation

10.1 Overview

The Reaction Wheel Attitude Control System (RWACS) is an effective method for controlling the orientation of nano satellites without using propellant. It operates on the principle of conservation of angular momentum, where variation in reaction wheel speed produces an equal and opposite rotation of the satellite. This enables precise and continuous attitude control required for communication alignment and payload orientation.

10.2 System Implementation

The RWACS was implemented through integrated mechanical design, modelling, and control system development. The reaction wheel assembly — including flywheel, motor housing, and mounting structure — was designed using CAD software with emphasis on balance, compactness, and vibration reduction.

System parameters such as moment of inertia, required torque, and angular velocity limits were calculated analytically. A dynamic model based on rigid body dynamics was developed to describe the interaction between the satellite and the reaction wheels. The 3D model was imported into MSC ADAMS for dynamic simulation, where multiple iterations were performed to optimize system performance. Structural validation was carried out in ANSYS through modal analysis, random vibration analysis, and housing strength evaluation to ensure reliability under operational and launch conditions.

10.3 Experimentation

Simulation-based experimentation was conducted to evaluate system performance under step inputs, disturbances, and multi-axis motion. The results indicate that the RWACS provides accurate, stable, and reliable attitude control suitable for nano satellite applications.

11. Results and Discussion

11.1 Modal Analysis of Flywheel

The natural frequencies and corresponding mode shapes of the flywheel were extracted. The analysis was performed as a pre-stressed modal analysis, accounting for the operational rotational effects of the flywheel. The first six natural frequencies were obtained in the range of 2000 Hz to 5000 Hz, which are significantly higher than the motor excitation frequency (200 Hz), thereby ensuring that resonance conditions are avoided during operation.

Key observation:

- Motor operating speed = 9000 RPM
- Excitation frequency = $9000 / 60 = 200$ Hz

Since none of the natural frequencies coincide with 200 Hz, the design is safe from resonance.

11.2 Random Vibration Analysis of Flywheel

The response of the flywheel under random vibration loading was evaluated using PSD analysis.

Result:

- Output: frequency vs. displacement PSD
- Maximum displacement observed within acceptable limits
- No peak observed near the excitation frequency (200 Hz)

The displacement results were evaluated at the mounting/interface locations, representing the most critical regions for structural integrity.

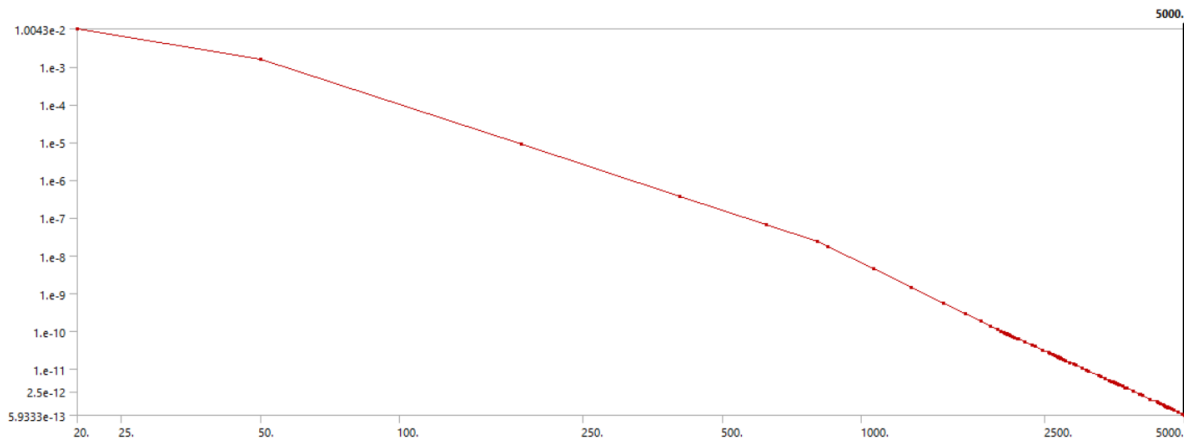


Fig. 11.1 — Frequency (X-axis) vs displacement (Y-axis) for the flywheel

Conclusion: the flywheel exhibits stable dynamic behaviour under random vibration loading and satisfies safety requirements.

11.3 Random Vibration Analysis of RWACS Housing

The RWACS housing was analyzed under the same PSD loading conditions.

Results:

- Displacement PSD response remains within safe limits
- No resonance observed near the excitation frequency
- Strain contours indicate uniform distribution without localized failure

The contour plots of normal elastic strain confirm that the structure can withstand the applied vibration loads without structural damage.

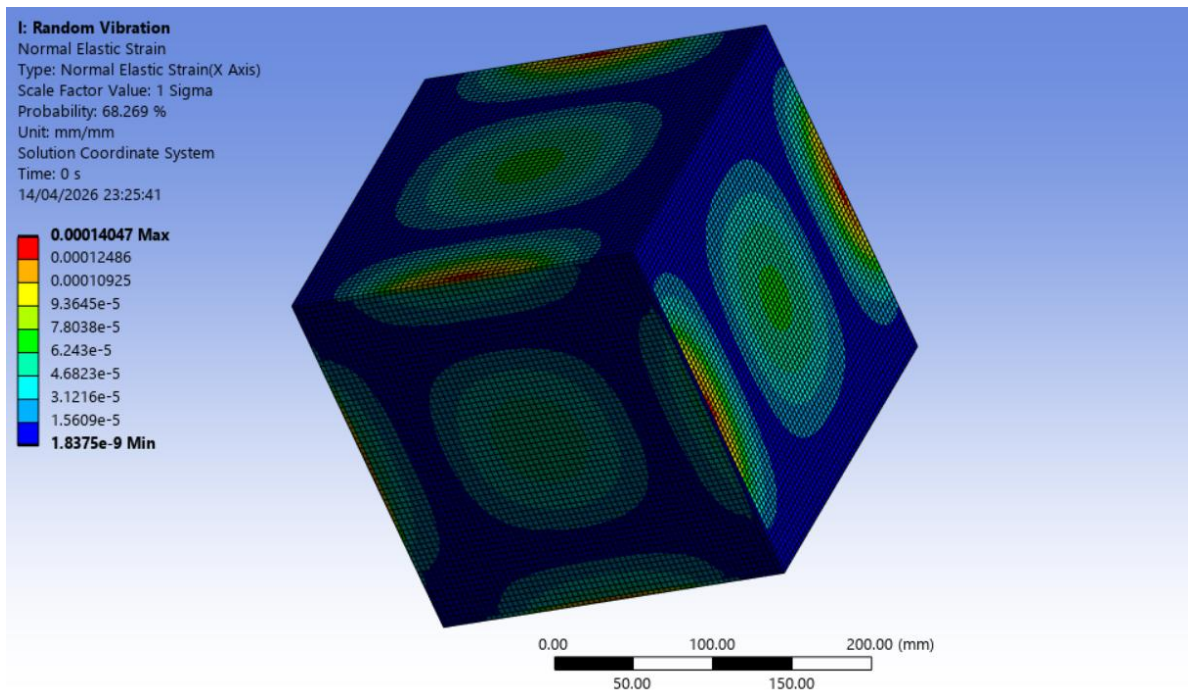


Fig. 11.2 — Random vibration response: contour of normal elastic strain (X-direction) under GEVS PSD loading

12. Conclusions and Scope of Future Work

The implementation and experimentation of the Reaction Wheel Attitude Control System (RWACS) demonstrate its effectiveness as a reliable and efficient solution for attitude control in nano satellites. By utilizing the principle of conservation of angular momentum, the system enables precise orientation control without the need for propellant, making it highly suitable for small satellite missions where mass and power constraints are critical.

The integration of mechanical design, analytical parameterization, and dynamic modelling ensured a strong theoretical foundation for system development. The use of advanced tools such as Creo Parametric, MSC ADAMS, and ANSYS enabled comprehensive design validation through multi-body dynamics simulation and structural analysis. Modal and random vibration analyses confirmed the structural integrity of the system under operational and launch conditions.

Simulation-based experimentation verified the system's ability to achieve stable and accurate attitude control under various conditions, including disturbances and multi-axis motion. Performance metrics such as settling time, overshoot, and steady-state error indicated satisfactory system response and robustness.

Overall, the study confirms that RWACS is a viable and scalable solution for nano satellite applications. The methodology adopted in this work provides a systematic approach for the design, analysis, and validation of attitude control systems, and can be extended to more advanced control strategies and real-time hardware implementation in future work.

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